A geometric/number theoretic proof of

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

<u>Theorem</u>: Let $p \in \mathbb{Z}^{\geq 0}$ be prime.

- $p \equiv 1 \mod 4 \implies p \in \mathbb{Z}[i]$ is the product of two conjugate primes, which are not associated;
- $p \equiv 3 \mod 4 \implies p \in \mathbb{Z}[i]$ stays prime.

Remark: Since $\mathbb{Z}[i]$ is a UFD, so factors are unique up to multiplication by units $\{\pm 1, \pm i\}$. We have $2 = (1+i)(1-i) = -i(1+i)^2$, and for the primes $p \equiv 1\ (4)$, whose factors are not associated by a unit, we have exactly 4 factorisations.

The number of lattice points inside a disk of radius R is approximately πR^2 as $R \to \infty$, ie $L_R = |D^2(R) \cap \mathbb{Z}^2| \approx \pi R^2$.

$\{(x,y) \in \mathbb{R}^2 : x^2+y^2=r^2\}$

Note that L_R is the sum of the number of lattice points on $S^1(r)$ for $r \leq R$, ie

$$L_R = \sum_{n \leq R^2} |S^1(\sqrt{n}) \cap \mathbb{Z}^2|.$$

if $(x,y) \in \mathbb{Z}^2 \cap D^2(R)$ then $(x,y) \in S^1(\sqrt{n})$ for some $n \in \mathbb{Z}$ since $x^2+y^2 \in \mathbb{Z}$

Now consider $\mathbb{R}^2 \cong \mathbb{C}$. Then the number of lattice points on $S^1(\sqrt{n})$ is the number of factorisations of n into conjugate factors in $\mathbb{Z}[i]$.

Let $n = 2^{\alpha_0} p_1^{\alpha_1} \cdots p_n^{\alpha_n} q_1^{\beta_1} \cdots q_m^{\beta_m}$, where $p_i \equiv 1 \mod 4$, $q_j \equiv 3 \mod 4$. Then p_i factors into two factors in exactly four ways, q_j does not factor, and 2 factors in two ways.

To find factorisations of n, we need to pick one factor from each factorisation. Hence,

- $\mathbb{Z}^2 \cap S^1(\sqrt{n}) = \emptyset$ if β_i is odd for some i,
- $|\mathbb{Z}^2 \cap S^1(\sqrt{n})|$ does not depend on α_0,
- for each p_i, $|\mathbb{Z}^2 \cap S^1(\sqrt{n})|$ increases by a factor of $\alpha_i + 1$.

factorisations into conj. factors only differ by a sign.

Now define $\chi : \mathbb{Z}^{\geq 0} \longrightarrow \{-1, 0, -1\}$ by

$$n \longmapsto \begin{cases} -1 & \text{if } n \equiv 3 \bmod 4 \\ 0 & \text{if } n \text{ is even} \\ 1 & \text{if } n \equiv 1 \bmod 4. \end{cases}$$

Note that χ is strongly multiplicative. By the above,

1 if β_j is even
0 o/w

$$\frac{1}{4}\left|\mathbb{Z}^2 \cap S^1(\sqrt{n})\right| = \left(\sum_{\gamma=0}^{\alpha_0} \underbrace{\chi(2^\gamma)}_{\substack{1 \text{ if } \gamma=0 \\ 0 \text{ o/w}}}\right) \prod_{i=1}^{n} \left(\sum_{\alpha=0}^{\alpha_i} \underbrace{\chi(p_i^{\alpha})}_{\text{always } 1}\right) \prod_{j=1}^{m} \left(\sum_{\beta=0}^{\beta_j} \underbrace{\chi(p_j^{\beta})}_{\pm 1}\right)$$

$$= \prod_{\substack{p \mid n \\ p \text{ prime}}} \left(\sum_{i=0}^{\mathrm{val}_p(n)} \chi(p^i)\right)$$

$$= \sum_{d \mid n} \chi(d).$$

Hence,

$$L_N = \overset{|S^1(0)|}{1} + 4 \sum_{n=1}^{N^2} \sum_{d \mid n} \chi(d)$$

each d appears as a divisor of appr. $\frac{N^2}{d}$ numbers below N^2

$$\approx 1 + 4 \sum_{d=1}^{N^2} \frac{N^2}{d} \chi(d)$$

$$\chi(d) = \begin{cases} 0 \text{ if } d \text{ is even} \\ \text{alternatingly } \pm 1 \text{ if } d \text{ is odd} \end{cases}$$

$$= 1 + 4N^2 \sum_{2d+1 \leq N^2} (-1)^d \frac{1}{2d+1}.$$

From $L_N \approx \pi N^2$, we get

$$\pi \approx \frac{1}{N^2} + 4\left(1 - \frac{1}{3} + \frac{1}{5} - \ldots \pm \frac{1}{\underset{\max\{2d+1 \leq N^2\}}{2d+1}}\right),$$

which becomes exact in the limit as $N \to \infty$.